higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 3

22 FEBRUARY 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 1.1.1

$$V_{cone} = \frac{1}{3}\pi r^2 \times \perp h$$
$$= \frac{1}{3}\pi(5^2)(8)\checkmark$$
$$= 209,439 \text{ cm}^3\checkmark$$

(2)

1.1.2

$$l = \sqrt{h^2 + r^2}$$
$$= \sqrt{8^2 + 5^2}$$
$$= 9,434\,\text{cm}\checkmark$$

Area of curved surface of a cone $= \pi r l$
$= \pi(5)(9,434)\checkmark$
$= 148,189\,\text{cm}^2\checkmark$

(3)

1.2 1.2.1

$$V_{marble} = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3}\pi.2^3\checkmark$$
$$= 33,510 \text{ cm}^3\checkmark$$

(2)

1.2.2

$$V_{water} = (20\times 10)\times 15\checkmark$$
$$= 3000 \text{ cm}^3\checkmark$$

(2)

1.2.3

Displacement = 2000 cm^3✓

$$\text{No. of marbles} = \frac{2000}{33,51}\checkmark$$
$$= 60\checkmark$$

(3)

1.3 1.3.1

$$AB = \sqrt{\left(1-(-2)\right)^2 + \left(4-(-2)\right)^2}\checkmark$$
$$= \sqrt{45}\checkmark$$
$$= 3\sqrt{5} \quad \text{or} \quad 6,708\checkmark$$

(3)

1.3.2

$$M\left(\frac{1-2}{2};\frac{4-2}{2}\right)$$
$$= \left(-\frac{1}{2};1\right)\checkmark\checkmark$$

(2)

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1.3.3
$$mBC = \frac{1-(-2)}{4-(-2)}\checkmark$$
$$= \frac{1}{2}\checkmark$$
$$mAD = -2\checkmark$$
$$AD: y - y_1 = m(x - x_1)$$
$$y - 4 = -2(x-1)\checkmark$$
$$y = -2x + 6 \qquad (5)$$

1.3.4
$$\theta = 180^0 - \tan^{-1}(2)\checkmark\checkmark$$
$$= 116,565^0 \qquad (3)$$
$$\checkmark$$
[25]

QUESTION 2

2.1
$$\frac{\sin^2 240°}{\cos^2 120°} + \sin 330°.\tan 225°$$
$$= \frac{(-\sin^2 60^0)}{(-\cos^2 60^0)} + (-\sin 30^0)(\tan 45^0)\checkmark$$
$$= \frac{\left(-\frac{\sqrt{3}}{2}\right)^2}{\left(-\frac{1}{2}\right)^2} - \frac{1}{2}(1)\checkmark$$
$$= 3 - \frac{1}{2} \quad \checkmark$$
$$= \frac{5}{2}\checkmark \qquad (4)$$

2.2
$$\text{LHS} = \frac{8}{1-\cos^2 A} - \frac{4}{1+\cos A}$$
$$= \frac{8}{(1-\cos A)(1+\cos A)} - \frac{4}{1+\cos A}\checkmark$$
$$= \frac{8-4(1-\cos A)}{(1-\cos A)(1+\cos A)}\checkmark$$
$$= \frac{8-4+4\cos A}{(1-\cos A)(1+\cos A)}\checkmark$$
$$= \frac{4+4\cos A}{(1-\cos A)(1+\cos A)}$$
$$= \frac{4(1+\cos A)}{(1-\cos A)(1+\cos A)}\checkmark$$
$$= \frac{4}{(1-\cos A)}\checkmark$$

$$= \text{RHS} \qquad (5)$$

2.3
$$2\tan(\theta+10°) = 4{,}162$$
$$\tan(\theta+10°) = 2{,}081$$
$\theta+10° = \tan^{-1}(2{,}081)\checkmark$ or $\theta+10° = 180° + \tan^{-1}(2{,}081)\checkmark$

θ+10°=64,334° $\theta+10^0 = 180° + 64{,}334°$

θ=54,334°✓ ✓ θ=244,334°✓ (5)

2.4
$$\frac{\text{PS}}{7{,}3} = \sin 63°\checkmark$$
$$\text{PS} = 7{,}3\sin 63°$$
$$= 6{,}504\checkmark$$
$$\frac{\text{SR}}{6{,}504} = \tan 49°\checkmark$$
$$\text{SR} = 6{,}504\tan 49°$$
$$= 7{,}482\checkmark \qquad (4)$$

2.5 2.5.1
$$\frac{\sin 25°}{28} = \frac{\sin 100°}{\text{AC}}\checkmark$$
$$AC = \frac{28\sin 100°}{\sin 25°}\checkmark$$
$$= 65 \text{ m} \quad \checkmark \qquad (3)$$

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2.5.2 $65^2 = 60^2 + BC^2 - 2(60)(\text{BC})\cos 80$✓

$4225 = 3600 + \text{BC}^2 - 20,838\,\text{BC}$

$\text{BC}^2 - 20,838\,\text{BC} - 625 = 0$✓

$\text{BC} = \dfrac{20,838 \pm \sqrt{434,222 + 2500}}{2}$✓

$= 37,503$ m✓ (4)

[25]

QUESTION 3

3.1 3.1.1

$Q_{2\,position} = \dfrac{1}{2}(24+1)$

$= 12,5$✓

$Q_2 = P_{12} + 0,5(P_{13} - P_{12})$

$= 43 + 0,5(45 - 43)$

$= 44$✓ (2)

3.1.2

$Q_{1\,posit} = \dfrac{1}{4}(25)$

$= 6,25$✓

$Q_1 = P_6 + 0,25(P_7 - P_6)$

$= 36 + 0,25(36 - 36)$

$= 36$✓

$Q_{3\,posit} = \dfrac{3}{4}(25)$

$= 18,75$✓

$Q_3 = P_{18} + 0,75(P_{19} - P_{18})$

$= 50 + 0,75(52 - 50)$

$= 51,5$✓

$Q_i = 15,5$✓ (5)

3.1.3 Upper fence $= 51,5 + 1,5(15,5)$✓

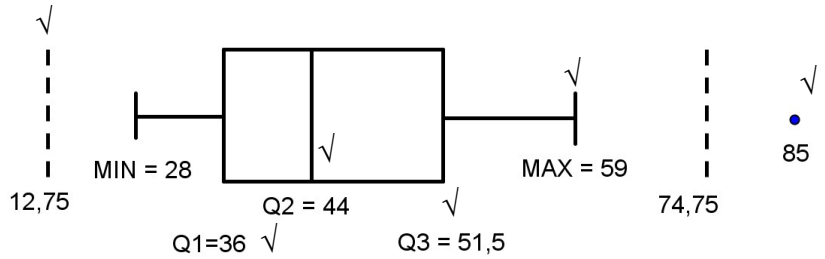
$= 74,75$✓

Lower fence $= 36 - 1,5(15,5)$✓

$= 12,75$✓ (4)

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3.1.4



(6 x ½) (3)

3.2 3.2.1

(a) 375
(b) 43
(c) 861
(d) 84
(e) 540
(f) 104
(g) 1056
(h) 136
(i) 1014
(j) 162
(k) 810
(l) 180

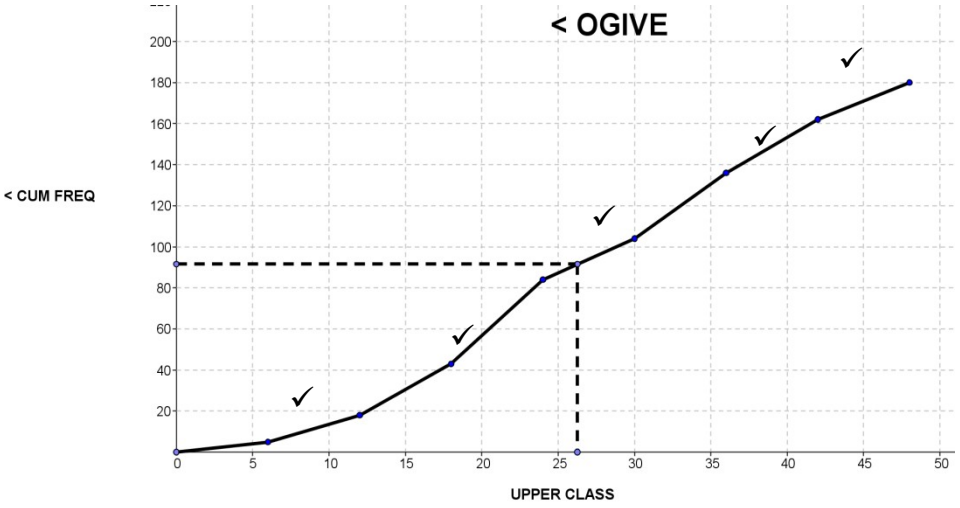
(12 x ½) (6)

3.2.2

$$\bar{x} = \frac{\sum f_1 x_1}{n} = \frac{4788}{180} \checkmark\checkmark\checkmark$$
$$= 26,6$$

(3)

3.2.3 **ADDENDUM**



(5)

3.2.4 26✓✓ (2)
[30]

QUESTION 4

4.1

4.1.1 +R1 800
4.1.2 R6 100
4.1.3 R1 775
4.1.4 R7 200
4.1.5 -R700
4.1.6 R1 775

(6 x 1) (6)

4.2 Bank

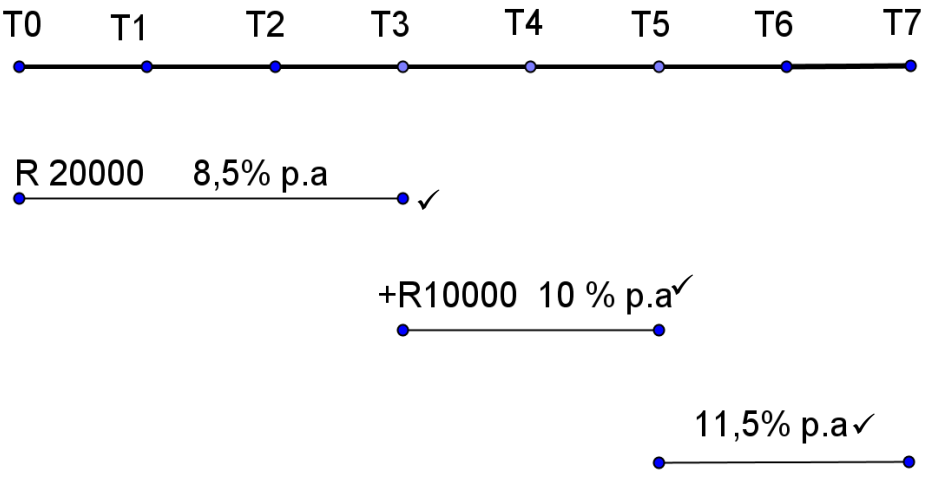
$$A = P(1+i)^n$$
$$= R210000\left(1+\frac{12}{1200}\right)^{60}$$
$$= R381506,31 \quad ✓✓$$

Uncle

$$A = P(1+i.n)$$
$$= R210000\left(1+\frac{12,5}{100}\times 5\right)$$
$$= R341250. \quad ✓✓$$

Saving = R40 256,31✓ (5)

4.3 4.3.1



(3)

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4.3.2

$$A_3 = 20000\left(1+\frac{8,5}{100}(3)\right)\checkmark$$

$$= R25100\checkmark$$

$$A_5 = (25100+10000)\left(1+\frac{10}{100}\right)^2\checkmark$$

$$= 42471\checkmark$$

$$A_7 = 42471\left(1+\frac{11,5}{200}\right)^4\checkmark$$

$$= 53114,64\checkmark$$

(6)
[20]

TOTAL: 100

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